

Mid-Mount Ladder Workgroup

Equipment Evaluation, Procurement Analysis, and Final Selection

Report – Q1 2026

MBFD Support Services

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i Note

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1 Executive Summary

In February 2026, the Miami Beach Fire Department convened a specialized workgroup to evaluate, rank, and select loose equipment for integration into a new Pierce Velocity 100-foot Ascendant mid-mount aerial tower. The workgroup conducted structured, multi-day hands-on evaluations of battery-operated extrication tools, rotary cut-off saws, vehicle stabilization systems, and specialty breaching tools from six manufacturers across fourteen product lines.

The evaluation dataset was forensically reconstructed to remove cost-based scoring influences. The original five-pillar model—which included an “Affordability” dimension—was corrected to a four-pillar framework reflecting pure operational performance: Capability, Usability, Maintainability, and Deployability. All scores reported in this document reflect the corrected methodology.

Key Findings:

- **Holmatro (Penttheon Series)** achieved the highest corrected brand average among extrication manufacturers at **90.72**, leading all four scoring dimensions. The Holmatro PSP40 spreader earned the single highest corrected tool score in the evaluation at **92.02**.
- **Hurst (eDRAULIC E3 Series)** ranked second with a corrected brand average of **86.53**, demonstrating competitive Capability scores (87.56) and superior saltwater protection (IP58).
- The **DeWalt 12-inch POWERSHIFT cut-off saw** dominated the rotary saw category with a corrected score of **91.25**, outperforming 14-inch competitors by more than 26 points.

- **Holmatro V-Struts** led vehicle stabilization at **87.28** with 15-second auto-lock deployment.

Final Workgroup Determination:

Despite Holmatro’s evaluation leadership, the workgroup selected the **Hurst E3 CAPTIUM platform** (SP 777 E3 spreader, S 789 E3 cutter, CR 522 E3 ram) as the primary frontline extrication system. This determination reflects operational factors beyond pure score ranking—including IP58 saltwater certification critical for Miami Beach’s coastal environment, ecosystem compatibility with the Hurst M40 heavy spreader assigned to the battalion chief’s vehicle, and Captium cloud-based fleet management integration. This divergence between top-scoring brand and final-selected brand is documented transparently throughout this report.

The complete recommended equipment package includes the Hurst E3 extrication suite, DeWalt 12-inch cut-off saw, DeWalt 18-inch chainsaw, four Holmatro V-Struts, the Holmatro T1 on trial basis as a rabbit tool replacement, and the Hurst M40 as a supplemental heavy extrication asset on the 300 vehicle.

2 Workgroup Background and Purpose

2.1 Origin and Command Directive

The Mid-Mount Ladder Workgroup was established by MBFD command to address a fundamental operational challenge: the department’s acquisition of a new Pierce Velocity 100-foot Ascendant aerial tower with a mid-mount configuration required a complete reassessment of loose equipment stowage, tool selection, and deployment doctrine.

Unlike rear-mount aerial platforms, a mid-mount configuration introduces asymmetrical compartment geometry that precludes direct transfer of existing equipment layouts. The workgroup was tasked with evaluating the commercial market, conducting structured hands-on testing, and delivering a consensus-driven “Top 3” finalist list for procurement authorization.

2.2 Command Structure

The workgroup operated under clear decision-making authority:

- **Consensus-driven model:** All configuration decisions were driven by workgroup vote
- **12 active workgroup members** serving as tactical decision-makers
- **Support Services** provided logistics, intelligence, and data analysis
- **Health & Safety Committee** held final procurement authorization
- **Fire Chief Digna Abello** served as reviewing authority

2.3 Day 1 Objectives

The inaugural session established the following priorities:

1. **Evaluate, rank, and select** loose equipment across all frontline operational categories
2. **Deliver a finalized procurement recommendation** to the Health & Safety Committee
3. **Assess platform ecosystem strategy**—specifically whether to commit exclusively to the DeWalt battery platform to match the truck’s existing charging infrastructure
4. **Optimize for narrow-profile tools** compatible with asymmetrical compartment constraints
5. **Address the saltwater/coastal exposure factor** in tool selection criteria

The workgroup identified four primary evaluation criteria from the outset: **Cost, Interoperability, Serviceability, and Lifecycle**—which were later refined into the structured four-dimension scoring framework.

3 Operational Context

3.1 The Mid-Mount Platform

The subject apparatus—Pierce Velocity 100-foot Ascendant aerial tower, Job #41559AD—presents specific engineering constraints that shaped the workgroup’s evaluation priorities:

- **TAK-4 independent suspension** with 24,880 lb rating and low center of gravity
- **2,000 GPM water output pump** supporting high-flow aerial and handline operations
- **Overall length of 528.50 inches** (44 feet, 0.5 inches)
- **Mid-mount turntable placement** displacing traditional compartment volumes

3.2 The Compartment Challenge: Asymmetrical Storage

The most consequential platform constraint identified on Day 1 was the asymmetrical compartment architecture inherent to the mid-mount design:

Side	Configuration	Implication
Driver Side (L Zones)	Full height, full depth	Primary bulk storage for fans, saws, rescue kits
Officer Side (R Zones)	Full height, <i>reduced depth</i>	Torque box intrusion limits stowage depth

The workgroup characterized this as “asymmetrical warfare”—the officer-side compartments cannot accommodate the same tool profiles as the driver side. This constraint eliminated the possibility of copying equipment layouts from existing apparatus and mandated optimization for narrow-profile tools across multiple categories.

Strategic implication: The department could not simply replicate existing configurations. Every tool selection required evaluation against the physical geometry of both compartment zones.

3.3 Coastal and Environmental Factors

Miami Beach’s barrier-island geography imposes environmental requirements on rescue equipment that inland departments do not face:

- **Saltwater exposure:** Proximity to the Atlantic Ocean and Biscayne Bay subjects equipment to persistent salt air and potential submersion scenarios
- **IP rating significance:** Ingress Protection ratings became a differentiating evaluation factor, with IP58 (saltwater-rated) tools receiving operational preference over IP54/IP57 alternatives
- **Corrosion lifecycle:** Equipment lifecycle projections must account for accelerated metallic degradation in marine environments

3.4 Platform Ecosystem Strategy

A foundational strategic question raised on Day 1 was whether MBFD should commit exclusively to the DeWalt battery platform to match the truck's integrated charging infrastructure. The workgroup identified this as a decision point across three categories:

- **Extrication tools:** Proprietary battery ecosystems (Hurst eDRAULIC, Holmatro Pentheon) vs. COTS platforms (Amkus on DeWalt 60V)
- **Forcible entry tools:** Battery-powered vs. manual hydraulic options
- **Saws:** Multiple competing battery architectures (DeWalt POWERSHIFT, Makita XGT, Husqvarna 94V)

The workgroup ultimately adopted a **hybrid ecosystem strategy**—selecting brand-specific proprietary batteries for primary extrication tools (where performance requirements dictate the power platform) while adopting COTS DeWalt batteries for secondary tools including saws and chainsaws.

4 Evaluation Framework and Methodology

4.1 Original Scoring Model

The workgroup's initial evaluation framework employed a five-dimension scoring model applied across all candidate products during hands-on testing sessions. Each dimension was scored on a 0–100 scale. The original five pillars included:

1. **Capability** — Raw performance: force output, cutting capacity, spreading distance, operational power
2. **Usability** — Ergonomic design, control interface, operator comfort, learning curve
3. **Maintainability** — Service requirements, durability, component accessibility, field-repairability
4. **Deployability** — Storage footprint, activation speed, weight, cordless readiness, compartment fit
5. **Affordability** — Cost-based considerations

4.2 Analytical Correction: Removal of Affordability

During post-evaluation data analysis, the Affordability dimension was identified as analytically problematic. Cost-based scoring introduced variability that did not reflect the pure operational performance of the tools under evaluation. Procurement cost is best evaluated independently through a separate budgetary process rather than blended into a composite operational performance score.

The scoring model was therefore corrected to a **four-pillar framework**:

$$\text{Corrected Overall Score} = f(\text{Capability, Usability, Maintainability, Deployability})$$

All scores presented in this report—brand averages, individual tool scores, and category rankings—reflect this corrected methodology with Affordability permanently removed.

The dashboard header for the reconstructed dataset states explicitly: *“Cost-based influences (‘Affordability’) have been permanently removed to reflect pure operational capability, usability, maintainability, and deployability.”*

4.3 Extrication Brand Scoring Standardization

Brand-level averages for extrication manufacturers were computed **exclusively from the performance of the mandatory frontline tactical triad**: the 32-inch spreader, cutter, and ram platforms. This standardization ensured that:

- **The Hurst M40** (40-inch heavy spreader) was excluded from the Hurst baseline brand average, as its specialized heavy-rescue profile would artificially depress the brand's frontline score
- **The Holmatro T1** (forcible entry tool) was excluded from the Holmatro baseline brand average, as its manual-hydraulic, multi-function role operates outside standard extrication parameters
- Both tools were isolated into **standalone specialty analysis**, evaluated on their own merits rather than distorting frontline brand comparisons

This approach preserves the analytical integrity of brand-to-brand frontline comparisons while ensuring specialty tools receive appropriate independent assessment.

4.4 Scoring Classification

The dashboard employs a four-tier classification system:

Tier	Score Range	Classification
	90+	Elite
	80–89	Highly Capable
	70–79	Acceptable
	< 70	Deficient

5 Finalist Evaluation Results

5.1 Extrication Brand Overall Performance (Corrected)

Four extrication manufacturers advanced to the final evaluation. Brand averages are derived strictly from the frontline triad (32-inch spreader, cutter, ram):

Table 1: Corrected brand-level averages for extrication manufacturers.

Rank	Brand	Corrected Avg	Capability	Usability	Maintainability	Deployability
#1	Holmatro	90.72	89.41	93.99	89.82	89.69
#2	Hurst	86.53	87.56	85.76	82.58	82.74
#3	TNT	75.12	74.67	78.38	65.78	81.67
#4	Amkus	73.56	84.21	66.98	69.14	73.89

Holmatro achieved Elite classification (90+) while Hurst achieved Highly Capable (80–89). TNT and Amkus fell into the Acceptable range.

The most significant dimension gap between the top two brands occurred in **Usability** (+8.23 points favoring Holmatro), attributable to the Pentheon Series' 360-degree inline control handle, On-Tool Charging system, and Stepless Speed Maximization technology. The narrowest gap appeared in **Capability** (+1.85), confirming that both platforms deliver competitive raw performance.

5.2 Frontline Extrication Tools: Individual Corrected Scores

Table 2: Frontline extrication tool rankings by corrected score.

Tool Category	Product Model	Manufacturer	Corrected Score
Spreader	PSP40 (32-inch)	Holmatro	92.02
Ram	PRA40	Holmatro	91.29
Spreader	SP 777 E3 (32-inch)	Hurst	90.99
Cutter	PCU30CL	Holmatro	88.86
Ram	CR 522 E3	Hurst	86.02
Cutter	S 789 E3	Hurst	82.57

5.2.1 Spreader Comparison

The Holmatro PSP40 achieved the **highest individual score in the entire evaluation** at 92.02, leading the Hurst SP 777 E3 (90.99) by 1.03 points. Both earned Elite classification. The PSP40's advantage derived primarily from its 360-degree inline control handle and Extreme Grip Spreader Tips, while the Hurst SP 777 E3 countered with substantially higher maximum spreading force (600 kN vs. 280 kN), wider 32-inch spreading distance (vs. 28.5 inches), and IP58 watertight design rated for saltwater operations.

5.2.2 Cutter Comparison

The Holmatro PCU30CL led the Hurst S 789 E3 by **6.29 points** (88.86 vs. 82.57)—the largest gap among the three frontline categories. The PCU30CL's 30-degree inclined jaw design maximizes working space between the tool and the vehicle, enabling safer and faster cutting without repositioning. The i-Bolt construction eliminates the need to retighten the central bolt after each use. The Hurst S 789 E3 offers a larger cutting opening of 8.07 inches (205 mm) vs. 6.7 inches (170.2 mm), but tracked lower in ergonomic usability under continuous load.

5.2.3 Ram Comparison

The Holmatro PRA40 dominated ram evaluations with a **91.29** corrected score versus the Hurst CR 522 E3 at **86.02** ($\Delta = 5.27$). The PRA40 achieved a 95.00 in operator Usability. Its integrated laser pointer provides first-time-right positioning, and at 31.1 lbs with a retracted length of only 15.2 inches, it is particularly suited to mid-mount compartment constraints. The Hurst CR 522 E3 counters with a telescopic dual-piston design offering 34.5 inches of overall stroke and IP58 watertight rating for saltwater deployment.

5.3 Rotary Cut-Off Saws

Table 3: Rotary cut-off saw corrected rankings.

Rank	Product Details	Manufacturer	Corrected Score
#1	12-inch (DCPS612AG2)	DeWalt	91.25
#2	14-inch (GEC01PL4)	Makita	64.83
#3	14-inch (K1 Pace)	Husqvarna	61.78

The DeWalt 12-inch POWERSHIFT platform established categorical dominance with a 91.25 corrected score—outperforming the Makita by **26.42 points** and the Husqvarna by **29.47 points**. Despite the

smaller blade diameter, the DeWalt’s gear-driven power delivery (not belt-driven), 3-second electric brake, and superior ergonomics rendered the 14-inch competitors operationally inferior in workgroup testing.

Both 14-inch saws received Deficient classification (< 70), with evaluators noting substantial penalties in Deployability and Usability due to excessive bulk, balance issues, and slower power spin-up times. The Makita GEC01PL4 offers power equivalent to a 75.6cc gas saw at 5,300 RPM, while the Husqvarna K1 Pace operates at 3,400 RPM—but their raw specifications did not translate to superior operational performance in hands-on evaluation.

5.4 Vehicle Stabilization

Table 4: Vehicle stabilization system rankings.

Rank	Product System	Manufacturer	Baseline Score
#1	V-Strut (Auto-locking)	Holmatro	87.28
#2	OmniShore (Pneumatic)	Holmatro	85.87
#3	Strut Driver (Mechanical)	Paratech	76.13

The Holmatro V-Strut’s auto-lock mechanism enables deployment in approximately 15 seconds—a single pull-and-lock movement with no separate locking operation required. At 15.87 lbs (7.2 kg), it is the lightest system evaluated. The integrated package combines the extending strut body, serrated multi-purpose head, non-slip base plate, and tensioning belt with hook and ratchet mechanism, eliminating loose components that can be lost on scene.

The Holmatro OmniShore system offers the most versatile capability set—supporting vehicle, trench, structural, and high-directional shoring from 28 cm to 5.2 m—but a documented deal-breaker concern was noted during the workgroup’s itinerary review. The Paratech StrutDriver provides 6,000 lbs lifting capacity with aircraft-grade aluminum construction but scored lowest in the category at 76.13.

6 Specialty Tool Considerations

The workgroup isolated two tools for standalone analysis. Due to their unique engineering, dimensions, and tactical deployment profiles, including them in standard frontline brand averages would produce mathematically inaccurate capability assessments.

6.1 Holmatro T1 — Multi-Purpose Forcible Entry Tool

Corrected Score: 82.23 (Highly Capable)

Table 5: Holmatro T1 key specifications.

Specification	Value
Weight	17.0 lbs (7.7 kg)
Spreading Force	33.0 kN (7,419 lbf)
Cutting Force	139.0 kN (31,248 lbf)
Maximum Spreading Distance	5.0 in (128 mm)
Cutting Opening	1.1 in (29 mm)
Power Source	Manual hydraulic (2-stage hand pump)

The T1 is a six-function-in-one tool: cut, wedge, ram, spread, hammer, and lift. It operates as a fully self-contained device—no batteries, no external pump, no hoses. Application of 30 kg (66 lbs) of manual force on the pump rod generates up to 14.2 tons of hydraulic cutting force and 3.4 tons of spreading force.

The workgroup evaluated the T1 specifically as a **potential replacement for the traditional rabbit tool** (halligan bar and flathead axe combination). The T1's detachable wedge enables single-operator door breaching—a significant advantage over the rabbit tool's traditional two-person requirement. At 17 lbs, it consolidates six functions into a weight comparable to married irons while adding hydraulic cutting capability absent from traditional forcible entry kits.

Workgroup consensus strongly recommends the T1 for rapid entry teams. Documented deal-breaker concerns were noted during evaluation, and the tool has been authorized for **trial deployment** rather than immediate full adoption.

6.2 Hurst M40 — 40-Inch Heavy Spreader

Corrected Score: 78.80 (Acceptable) | Capability: 83.39 | Usability: 79.29

The M40 provides 40-inch spreading distance for heavy rescue operations that exceed the parameters of standard 32-inch frontline spreaders. Its Capability score of 83.39 demonstrates substantial raw power, but its Usability score of **79.29—the lowest in the entire dataset**—reflects the immense physical footprint that severely limits standard operational handling.

The workgroup determined the M40 is **intended for heavy rescue apparatus and should not be deployed as a primary tool on standard frontline engines**. Within MBFD's fleet strategy, the M40 is selected as a supplemental asset for dedicated apparatus—specifically the 300 (battalion chief's vehicle)—providing heavy extrication capability at the command level that exceeds standard 32-inch spreader geometry.

7 Final Workgroup Determinations and Purchasing Recommendations

7.1 The Selection–Score Divergence

The most significant analytical finding in this report is the divergence between evaluation scoring leadership and the final workgroup selection for the primary extrication platform.

Holmatro (Pentheon Series) led every frontline scoring dimension and achieved the highest brand average (90.72), the highest individual tool score (PSP40 at 92.02), and Elite classification across the board. Under a purely score-driven procurement model, Holmatro would be the unambiguous selection.

The workgroup selected Hurst (eDRAULIC E3 CAPTIUM) as the primary extrication platform.

This determination was not an error, nor does it reflect a disregard for the evaluation data. The workgroup exercised its consensus-driven authority to weigh operational factors that the four-pillar corrected scoring model captures incompletely:

1. **Saltwater certification (IP58):** The Hurst E3 platform carries IP58 watertight certification rated for saltwater operations. The Holmatro Pentheon tools carry IP57. For a barrier-island department with Atlantic Ocean and Biscayne Bay exposure, the higher ingress protection rating represents a measurable operational advantage in corrosion resistance and submersion tolerance.

2. **Ecosystem interoperability with the M40:** The workgroup simultaneously selected the Hurst M40 heavy spreader for the 300 vehicle. Selecting Hurst for both the frontline triad and the supplemental heavy spreader consolidates battery platform, maintenance pipeline, vendor relationship, and training requirements onto a single manufacturer ecosystem.
3. **Captium cloud integration:** The E3 Connect platform with Wi-Fi data transfer to the Captium cloud provides fleet-management capabilities—tool usage tracking, battery health monitoring, and maintenance scheduling—that the Pentheon platform’s Bluetooth/MyHolmatro ecosystem does not replicate at the same fleet integration level.
4. **Capability parity:** The Capability dimension gap between Holmatro (89.41) and Hurst (87.56) is only 1.85 points. Raw performance is effectively at parity; the scoring divergence is driven primarily by Usability and Maintainability, where Holmatro’s ergonomic advantages are genuine but do not override the IP58 and ecosystem considerations for this specific department.

This divergence is documented here in the interest of analytical transparency. The evaluation data stands as presented; the final determination reflects the workgroup’s informed professional judgment integrating factors beyond the four-pillar score.

7.2 Final Selected Equipment Package

Table 6: Final workgroup equipment selections.

Category	Selected Equipment	Platform Assignment
Cut-Off Saw	DeWalt 12” Battery Cut-Off Saw (DCPS612AG2)	Frontline (L1/L3)
Chainsaw	DeWalt 18” Chainsaw (bullet chain + depth markings)	Frontline (L1/L3)
Stabilization	4 × Holmatro V-Struts	Frontline (L1/L3)
Spreader	Hurst SP 777 E3 Connect	Frontline — CAPTIUM Platform
Cutter	Hurst S 789 E3 Connect	Frontline — CAPTIUM Platform
Ram	Hurst CR 522 E3 Connect	Frontline — CAPTIUM Platform
Specialty Tool	Holmatro T1 (Trial Authorization)	Frontline — Rabbit Tool Replacement Candidate
Heavy Extrication	Hurst M40 + 2 batteries + charging station	300 (Battalion Chief Vehicle)
Future Addition	Lifting Struts (pending evaluation)	300 / Captain 5

8 Apparatus Deployment and Implementation Strategy

8.1 Tiered Deployment Model

The workgroup established a tiered deployment architecture ensuring full operational independence across all rescue scenarios without dependency on mutual aid.

8.1.1 Frontline Apparatus (L1 / L3)

Equipment	Deployment Role
Hurst SP 777 E3 Connect Spreader	Primary extrication spreading
Hurst S 789 E3 Connect Cutter	Primary extrication cutting
Hurst CR 522 E3 Connect Ram	Extension / displacement operations
4 × Holmatro V-Struts	Rapid vehicle stabilization
DeWalt 12" Battery Cut-Off Saw	Rapid entry / concrete-steel cutting
DeWalt 18" Chainsaw	Ventilation operations
Holmatro T1 (trial)	Multi-function access / forcible entry

Operational priority: Rapid deployment capability, compartment-optimized stowage, battery-independent operation across mixed ecosystems.

8.1.2 Command Level (300 / Captain 5)

Equipment	Deployment Role
Hurst M40 40" Spreader	Heavy extrication beyond standard parameters
2 × CAPTIUM Batteries	Extended power supply
Charging Station	Field recharging capability
Lifting Struts (future)	Vehicle lifting operations

Operational priority: Heavy rescue augmentation, command-level specialty capability for scenarios exceeding standard 32-inch spreader geometry—including large commercial vehicle extrication, structural collapse, and multi-vehicle incidents.

8.1.3 Heavy Rescue (E2 / Box Truck)

High-capacity rescue systems for extreme lifting and technical rescue scenarios.

Operational priority: Maximum force, specialist operations, structural rescue.

Strategic goal: Full operational independence across all rescue levels—ensuring MBFD units can handle any vehicle extrication scenario from routine motor vehicle collisions through heavy commercial entrapment without dependency on mutual aid resources.

9 Training and In-Service Plan

9.1 Train-the-Trainer Model

Workgroup members who participated in the evaluation process will serve as **subject matter experts (SMEs)** and lead departmental training through a phased implementation.

9.1.1 Phase 1 — Vendor Training

Workgroup members receive comprehensive manufacturer-led training on all selected equipment systems. Members achieve proficiency qualification on each platform before conducting internal training.

9.1.2 Phase 2 — Department-Wide Rollout

Department personnel are divided across shift groups. Each workgroup SME trains their assigned shift. **No overtime is required**—training is integrated into existing shift schedules, minimizing budget impact while ensuring complete departmental coverage.

9.1.3 Phase 3 — Deep Familiarization

All personnel complete training covering:

- Tool operation and safety protocols for each selected system
- Tactical application in vehicle extrication scenarios specific to the mid-mount platform
- Battery management and field charging procedures across both Hurst CAPTUM and DeWalt ecosystems
- V-Strut stabilization deployment procedures
- Integration of the T1 trial tool into existing forcible entry protocols
- M40 heavy spreader operation for personnel assigned to the 300

9.2 Ladder Truck In-Service Requirements

Prior to placing the mid-mount ladder truck into operational service, Support Services will ensure all assigned personnel complete:

Training Component	Requirement
Driving Operations	Full certification on mid-mount ladder configuration
Pumping Operations	Operational proficiency on all pump functions
Tactical Deployment	Extrication tool deployment from mid-mount compartments
Equipment Integration	Familiarization with all selected equipment as installed

All training components must be completed **before** the apparatus enters service.

10 Strategic Conclusion

The MBFD Mid-Mount Ladder Workgroup has completed a rigorous, data-driven evaluation of rescue equipment spanning six manufacturers, fourteen products, and four operational categories. The corrected four-pillar scoring framework—with Affordability permanently removed—provides command staff with an unambiguous performance baseline for current and future procurement decisions.

The final equipment package reflects a deliberate balance of quantitative evaluation data and operational judgment. Where the scoring data pointed clearly to a single solution (DeWalt for saws, Holmatro for stabilization), the workgroup followed the data directly. Where the top two extrication platforms demonstrated competitive performance with differentiated operational advantages, the workgroup exercised its consensus authority to select the platform best aligned with MBFD’s coastal operating environment, fleet integration strategy, and heavy-rescue interoperability requirements.

The workgroup remains active through full implementation of the selected equipment. Ongoing responsibilities include procurement tracking, delivery coordination, installation oversight, dashboard monitoring, and leadership of the scheduled follow-on evaluation session for lifting struts.

11 Appendices

11.1 Appendix A — Product Evaluation Data Summary

Table 7: Complete product evaluation data sorted by corrected score.

Product	Category	Manufacturer	Corrected Score
Holmatro PSP40 (32" Spreader)	Hydraulic Tool	Holmatro	92.02
DeWalt DCPS612AG2 (12" Saw)	Rescue Saw	DeWalt	91.25
Holmatro PRA40 (Ram)	Hydraulic Tool	Holmatro	91.29
Hurst SP 777 E3 (32" Spreader)	Hydraulic Tool	Hurst	90.99
Holmatro PCU30CL (Cutter)	Hydraulic Tool	Holmatro	88.86
Holmatro V-Strut	Stabilization	Holmatro	87.28
Hurst CR 522 E3 (Ram)	Hydraulic Tool	Hurst	86.02
Holmatro OmniShore	Stabilization	Holmatro	85.87
Hurst S 789 E3 (Cutter)	Hydraulic Tool	Hurst	82.57
Holmatro T1	Specialty	Holmatro	82.23
Hurst M40 (40" Spreader)	Specialty	Hurst	78.80
Paratech StrutDriver	Stabilization	Paratech	76.13
Makita GEC01PL4 (14" Saw)	Rescue Saw	Makita	64.83
Husqvarna K1 Pace (14" Saw)	Rescue Saw	Husqvarna	61.78

11.2 Appendix B — Data Source Notes

All numerical evaluation scores are sourced from the forensically reconstructed Executive Procurement Dashboard. Dashboard scores are designated TIER 1 authority—where any conflict exists between dashboard values and PDF-extracted text, the dashboard values govern.

Product specifications are sourced from manufacturer catalogs processed through the Docling document intelligence pipeline, cross-referenced against original vendor PDFs.

The corrected scoring methodology removes the Affordability dimension from all composite calculations. Original uncorrected scores are retained in archival records but are not presented in this report.

11.3 Appendix C — Scoring Key

- **90+** (Elite) — Exceeds all operational requirements with distinction
- **80–89** (Highly Capable) — Meets or exceeds operational requirements
- **70–79** (Acceptable) — Meets minimum operational requirements
- **< 70** (Deficient) — Falls below operational requirements in one or more critical dimensions